

services\analysis\trend_analyzer.py

```
from typing import Dict, List, Any
from datetime import datetime, timezone, timedelta
from collections import defaultdict
from services.data.api_client import api_client

def get_cve_trends_last_30_days() -> Dict[str, Any]:
    """Get CVE trends for last 30 days - API ONLY"""
    print("[CVE Trends] Fetching 30-day trends from API...")

    # Get recent vulnerability data from API
    cves = api_client.get_cves_last_30_days()

    # Initialize daily counting structure
    daily_counts = defaultdict(int)
    today = datetime.now(timezone.utc).date()

    # Pre-fill last 30 days with zero counts
    for i in range(30):
        day = today - timedelta(days=29-i)
        daily_counts[day.strftime('%Y-%m-%d')] = 0

    # Process each CVE to count by publication date
    for cve in cves:
        pub_date = cve.get('Published', '')
        if pub_date:
            try:
                if 'T' in pub_date:
                    dt = datetime.fromisoformat(pub_date.replace('Z', '+00:00'))
                else:
                    dt = datetime.strptime(pub_date[:10], '%Y-%m-%d')

                date_key = dt.date().strftime('%Y-%m-%d')
                if date_key in daily_counts:
                    daily_counts[date_key] += 1
            except:
                continue

    # Build chart-ready data structure
    sorted_dates = sorted(daily_counts.keys())
    result = {
        'labels': sorted_dates,
        'values': [daily_counts[date] for date in sorted_dates],
        'total_cves': len(cves),
        'date_range': {
            'start': sorted_dates[0] if sorted_dates else '',
            'end': sorted_dates[-1] if sorted_dates else ''
        }
    }

    print(f"[CVE Trends] Generated trends: {result['total_cves']} CVEs over {len(sorted_dates)} days")
    return result

def get_vulnerabilities_over_time_last_n_years(years=1) -> Dict[str, Any]:
    """Get vulnerability timeline - SUPPORTS MULTIPLE YEARS"""
    print(f"[Vulnerabilities Over Time] Requested data for {years} years")

    # Validate years parameter
    years = max(1, min(5, int(years))) # Limit between 1 and 5 years

    from services.data.data_processor import historical_loader

    # Get years to load based on parameter - THE KEY FIX!
    # REMOVED: current_year = datetime.now().year
    # REMOVED: years_to_load = [current_year] # CHANGED: Always only current year!

    # NEW CODE: Respect the years parameter
    current_year = datetime.now().year
    years_to_load = [current_year - i for i in range(years)]

    print(f"[Vulnerabilities Over Time] Loading years: {years_to_load}")
```

```

monthly_counts = defaultdict(int)
total_cves = 0

for year in years_to_load:
    year_cves = historical_loader.get_year_data(year)
    print(f"[Vulnerabilities Over Time] Processing {year}: {len(year_cves)} CVEs")
    total_cves += len(year_cves)

for cve in year_cves:
    pub_date = cve.get('Published', '')
    if pub_date:
        try:
            if 'T' in pub_date:
                dt = datetime.fromisoformat(pub_date.replace('Z', '+00:00'))
            else:
                dt = datetime.strptime(pub_date[:10], '%Y-%m-%d')
            month_key = dt.strftime('%Y-%m')
            monthly_counts[month_key] += 1
        except Exception as e:
            continue

# Clear cache after processing to save memory
if str(year) in historical_loader._year_cache:
    print(f"[Vulnerabilities Over Time] Clearing cache for {year} to save memory")
    del historical_loader._year_cache[str(year)]

sorted_months = sorted(monthly_counts.items())

result = {
    'labels': [item[0] for item in sorted_months],
    'values': [item[1] for item in sorted_months],
    'total_cves': total_cves,
    'months_covered': len(sorted_months),
    'raw_data': dict(monthly_counts),
    'years_included': years_to_load # Added this to track which years are included
}

print(f"[Vulnerabilities Over Time] Generated timeline: {result['total_cves']} CVEs
across {result['months_covered']} months")

return result

def get_filtered_historical_cves(year: int = None, month: int = None, day: int = None)
-> List[Dict[str, Any]]:
    """Get filtered CVEs from historical data - LAZY LOADING - MEMORY OPTIMIZED"""
    print(f"[Historical Filter] Getting filtered data for year={year}, month={month},
day={day}")

    # Year is required for historical filtering
    if not year:
        print("[Historical Filter] No year specified, returning empty list")
        return []

    # Lazy import to avoid circular dependencies
    from services.data.data_processor import historical_loader

    # Load specific year data for efficient filtering - LAZY LOAD
    year_cves = historical_loader.get_year_data(year)
    print(f"[Historical Filter] Loaded {len(year_cves)} CVEs for year {year}")

    # If no month specified, return all CVEs for the year
    if not month:
        print(f"[Historical Filter] No month filter, returning all {len(year_cves)} CVEs for
{year}")
        return year_cves

    # Apply hierarchical filtering: year ? month ? day
    filtered_cves = []
    for cve in year_cves:
        pub_date = cve.get('Published', '')
        if pub_date:
            try:
                if 'T' in pub_date:
                    dt = datetime.fromisoformat(pub_date.replace('Z', '+00:00'))
                else:
                    dt = datetime.strptime(pub_date[:10], '%Y-%m-%d')

```

```

# CRITICAL: Check BOTH year and month
if dt.year == year and dt.month == month:
    # If day is specified, check day too
    if day is None or dt.day == day:
        filtered_cves.append(cve)
    except Exception as e:
        continue

print(f"[Historical Filter] Filtered result: {len(filtered_cves)} CVEs for
{year}-{month:02d}" + (f"-{day:02d}" if day else ""))

# Debug: Show sample of filtered results
if filtered_cves:
    sample_count = min(3, len(filtered_cves))
    print(f"[Historical Filter] Sample of filtered CVEs:")
    for i in range(sample_count):
        cve = filtered_cves[i]
        print(f"    - {cve.get('ID', 'Unknown')}: {cve.get('Published', 'No date')}")

return filtered_cves

```